

Week 14 – Ship Your Own RAG Graded Mini Project 14.3

This is a graded mini project. You will build and deploy a public RAG bot over a document set you control. The goal is not “it works on my machine.” The goal is to open and use a public product.

Practice Context – From Workshop Bot to Public Product

You will be working from the Next.js RAG starter and turning it into a public, domain-specific application.

This project asks you to:

- Choose a useful corpus
- Build a retrieval system over it
- Expose it through a streaming chat interface
- Render citations
- Deploy it publicly
- Explain the trade-offs you made

Deliverables

You must submit:

- Deployed public URL
- Github repository link
- One-page reflection
- Optional demo questions

How to approach this mini-project

1. Set up the starter project and repo first.
2. Choose your corpus carefully before touching chunk size.
3. Add metadata that makes citations useful.
4. Test the bot with many real questions locally.

5. Tune chunking, prompt, tool description, and topK.
 6. Polish the UI before deploying.
 7. Test the production version in incognito.
 8. Write the reflection only after real usage and debugging.
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Key Requirements

Your submission must:

- Be publicly reachable on Vercel
 - Stream answers
 - Use retrieval as a tool call
 - Show retrieved sources
 - Avoid client-side secret leakage
 - Include a meaningful empty state
 - Be grounded in a real corpus you chose
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Recommended Timeline

A good weekly split is:

- **Day 1:** choose corpus and finish setup
- **Day 2:** seed, test, and iterate on retrieval
- **Day 3:** polish and deploy
- **Day 4:** reflection and final verification

Total expected effort:

- about **8–12 hours**
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Stretch Goals

Choose at most two, and do them well.

Examples:

- Source-PDF download from citation
 - /usage page
 - Suggested-prompt chips
 - Mobile-friendly layout
 - Multi-PDF support
 - Hybrid retrieval
 - Evaluation harness
 - Observability
 - Auth and rate limiting
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Troubleshooting Focus

Common issues include:

- Missing or misnamed production env vars
 - Vague tool description causing general-knowledge answers
 - Bad chunking causing irrelevant retrieval
 - Missing metadata causing empty sources panel
 - Cold-start or rate-limit issues after deployment
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Submission

Submit in the place specified by your facilitator:

- Deployed URL
- Repo link
- Reflection PDF
- Optional demo questions
- Optional stretch goals summary